State Estimation and Filtering for Robotic Perception

AA273

Gaussian RVs, Expectation, Mean, Covariance

Moments, Conditional Expectation



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

Announcements:

- HW 2 out Sun Jan 18, due Sun Jan 25
- HW 3 out Fri Jan 23, due Fri Jan 30
- Project topic ideas posted
- Group signup sheet posted (groups from 1 – 4 people)

Lecture 5:

- Multivariate Gaussian RVs
- Expectation, mean, covariance
- Conditional expectation
- Jointly Gaussian RVs

Multivariate Gaussian RVs

RV takes on vector values: $x \in \mathbb{R}^n$

Def. of multi. var. Gaussian:

$$p(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

Shorthand: $X \sim \mathcal{N}(\mu, \Sigma)$

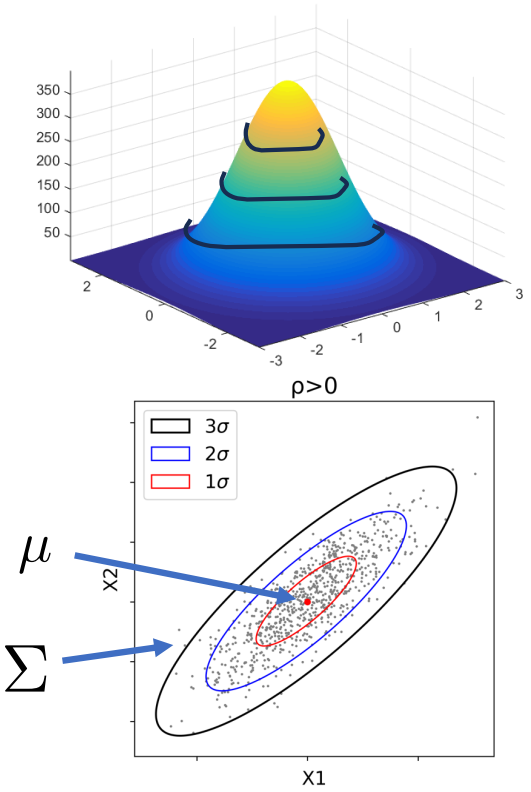
Parameterized by:

Mean: $\mu \in \mathbb{R}^n$

Covariance: $\Sigma \in \mathbb{R}^{n \times n}$

Symmetric, positive semi-definite

$\Sigma^T = \Sigma \qquad x^T \Sigma x \geq 0 \quad \forall x$



Confidence Ellipse

Ellipse *B* centered at mean s.t.

$$Pr(x \in \mathcal{B}(c)) = P(c)$$

Defined by a level set of the pdf:

$$\mathcal{B}(c) = \{x \mid \mathcal{N}(x) \geq c\}$$

Eigenvector, eigenvalue give direction and mag of principal axes

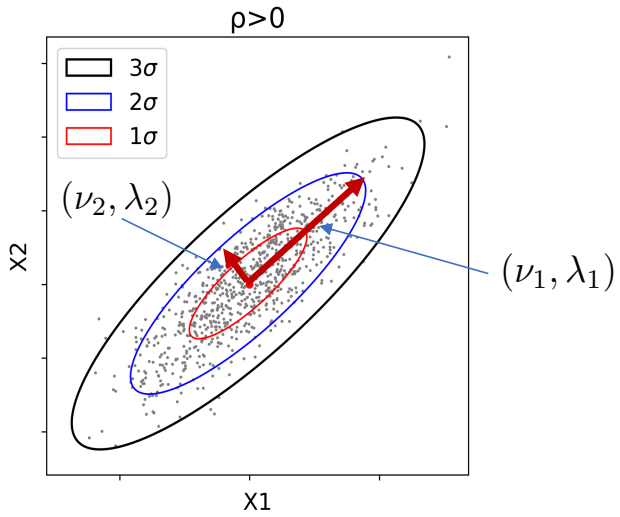
$$(\nu_i, \lambda_i) \quad \text{s.t.} \quad \lambda_i \nu_i = \Sigma \nu_i$$

How to get equation?

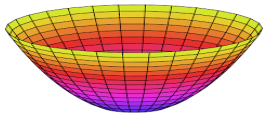
$$\mathcal{N}(x) = \eta e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} = c$$

$$(x-\mu)^T \Sigma^{-1}(x-\mu) = 2\log(\frac{\eta}{c})$$

$$(x-\mu)^T \Sigma^{-1}(x-\mu) = \kappa(c)$$

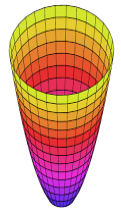


$\Sigma \uparrow \implies \Sigma^{-1} \downarrow \implies$



High uncertainty

$\Sigma \downarrow \implies \Sigma^{-1} \uparrow \implies$



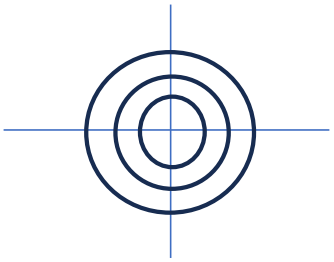
Low uncertainty

Standard Normal Dist.

$$X_s \sim \mathcal{N}(0, I)$$

$\mu = 0$ $\quad$ $\Sigma = I$

Level sets are circles centered at origin



Standard normal can be transformed into any Gaussian RV of same dim.

$$X = \Sigma^{\frac{1}{2}} X_s + \mu$$

gives $X \sim \mathcal{N}(\mu, \Sigma)$

Helpful for computing integrals in HW 2!

Any Gaussian RV can be transformed into Standard Normal of same dim.

given $X \sim \mathcal{N}(\mu, \Sigma)$

$$X_s = \Sigma^{-\frac{1}{2}}(X - \mu)$$

s.t. $X_s \sim \mathcal{N}(0, I)$

Aside: Martrix square root

$M^{\frac{1}{2}}$ matrix square root, defined s.t. $\left(M^{\frac{1}{2}}\right)^T M^{\frac{1}{2}} = M$

There are several to choose from. We'll see more details later.

Expectation of a RV

Given RV: $X \sim p(x),\ x \in \mathbb{R}^n$

Expectation of RV: $E[X] = \int x\, p(x)\, dx$ For discrete: $E[X] = \sum_x x\, p(x)$

Also called **mean** of X, with notation: $\mu_x := E[X]$

Expectation of a general function of X: $E[f(X)] := \int_x f(x)p(x)\, dx$

E.g., for f() affine: $f(X) = Ax + b$

$$E[AX + b] = \int_x (Ax + b)p(x)\, dx = A \int_x xp(x)\, dx + b$$

So expectation is a **linear operator:** $E[AX + b] = AE[X] + b$

Covariance

Covariance of RV:

Vector outer product

$$Cov(X) := E[(X - \mu_x)(X - \mu_x)^T] = \int_x (x - \mu_x)(x - \mu_x)^T p(x)\, dx$$

We use notation: $\Sigma_x := Cov(X)$

$$Cov(X) = \begin{bmatrix} Cov(X_1, X_1) & Cov(X_1, X_2) & \cdots & Cov(X_1, X_n) \\ \vdots & \vdots & \cdots & \vdots \\ Cov(X_n, X_1) & Cov(X_n, X_2) & \cdots & Cov(X_n, X_n) \end{bmatrix}$$

Matrix of scalar **cross covariances**:

$$Cov(X_i, X_j) := E[(X_i - \mu_{x_i})(X_j - \mu_{x_j})^T] = \int_{x_i} \int_{x_j} (x_i - \mu_{x_i})(x_j - \mu_{x_j})^T p(x_i, x_j)\, dx_i\, dx_j$$

Intuition: How often does X_i go up/down when X_j does.

Vector Cross-Covariance

For two RVs $X \sim p(x),\ x \in \mathbb{R}^n$ $\qquad$ $Y \sim p(y),\ y \in \mathbb{R}^m$

Cross covariance defined as:

$$\Sigma_{xy} = Cov(X, Y) := E[(X - \mu_x)(Y - \mu_y)^T] \in \mathbb{R}^{n \times m}$$

Or in Einstein notation:

$$Cov(X, Y) := E\left[[(X_i - \mu_{x_i})(Y_j - \mu_{y_j})]_{i,j}\right] \in \mathbb{R}^{n \times m}$$

Matrix of scalar integrals:

$$= \int_{x_i} \int_{y_j} (x_i - \mu_{x_i})(y_j - \mu_{y_j}) p(x_i, y_j)\, dx_i\, dy_j$$

Notice, given: $Z = (X, Y) \sim p(x, y)$ $\qquad$ $\Sigma_z = \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}$

Moments of RVs

Mean is called 1st moment: $\mu_x = E[X] \in \mathbb{R}^n$

Cov is called 2nd moment: $\Sigma_x = E[(X - \mu_x)(X - \mu_x)^T] \in \mathbb{R}^{n \times n}$

3rd moment (skew): $E\left[[(X_i - \mu_{x_i})(X_j - \mu_{x_j})(X_k - \mu_{x_k})]_{i,j,k}\right] \in \mathbb{R}^{n \times n \times n}$

4rh moment (kurtosis): $E\left[[(X_i - \mu_{x_i})(X_j - \mu_{x_j})(X_k - \mu_{x_k})(X_l - \mu_{x_l})]_{i,j,k,l}\right] \in \mathbb{R}^{n \times n \times n \times n}$

$\vdots$

Intuition: Characterize the shape of the distribution in a kind of polynomial expansion, like Fourier expansion.

Law of the Unconscious Statistician (LOTUS)

Given RV: $X \sim p(x)$

And another RV as function of the first: $Y = f(X)$

LOTUS: $E_y[Y] = E_x[f(X)]$

$$\int_y y p(y)\, dy = \int_x f(x) p(x)\, dx$$

Intuition: No need to explicitly find the dist. p(y), we can integrate using p(x) directly.

Proof of Standard Normal Eqns

Notes on the board.

Jointly Gaussian RVs

Notes on the board.